WENJUN YU

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EDUCATION

Hong Kong Baptist University **Sep 2025 – Present**
Ph.D. student in Computer Science
Supervisor: Amelie Chi Zhou

Dalian University of Technology (985 & 211) **Sep 2019 – July 2023**
B.S. in Artificial Intelligence

RESEARCH INTEREST

My research interests lie broadly in DLRMs training/serving optimization, generative Recommender system acceleration and GPU Virtualization techniques.

PUBLICATIONS

When KV Meets Embeddings: Dynamic GPU Memory Allocation for Accelerating Generative Recommender Serving
Wenjun Yu, Amelie Chi Zhou, Shuguang Han (work with Alibaba – Xianyu)
(SC 2026, CCF-A 19.2% acceptance rate) International Conference for High Performance Computing, Networking, Storage, and Analysis
Generative Recommender inference faces a dual memory bottleneck absent in LLM serving: embedding hot caches and KV caches compete for the same limited GPU HBM, yet existing systems optimize them in isolation. We present RACER, which jointly reallocates HBM between the two caches at runtime and routes requests via a joint EMB–KV-aware scheduler, reducing P99 latency by 24–38% while maintaining $\geq$ 93.5% SLO satisfaction across production workload regimes.

Near-Zero-Overhead Freshness for Recommendation Systems via Inference-Side Model Updates
Wenjun Yu, Sitian Chen, Amelie Chi Zhou, Cheng Chen (work with ByteDance – Douyin)
(HPCA 2026, CCF-A, Top-4 Computer Architecture Conference, 19.7% acceptance rate)
IEEE International Symposium on High-Performance Computer Architecture
Updating online DLRMs is challenging due to their large model size. We address this by leveraging idle online CPU resources to perform continuous updates, thereby maintaining model freshness. Our approach employs dynamic rank adaptation and NUMA-aware resource scheduling to deliver these updates, significantly improving recommendation accuracy without impacting inference latency.

WORK EXPERIENCE

Research Assistant **Feb. 2025 – Aug. 2025**
Hong Kong Baptist University, Bytedance Douyin *Hong Kong, China*
• Designed a LoRA-based online update mechanism for recommendation models, reducing communication overhead during model updates. Work is published in HPCA 2026.

GPU Virtualization System Engineer **Aug. 2023 – Feb. 2025**
VirtAi Technology *Beijing, China*
• Design a playback-based hot migration module for NCCL-based deep learning programs, which allows deep learning tasks to be migrated to different GPUs within a single machine without interrupting the training process
• Design a GPU computing resource control module, it hooks the CUDA launch kernel APIs, then monitors and controls the CUDA kernel rate for deep learning jobs to balance GPU computing resources or set computing resource priorities for different tasks in the OS layer
• Design a fault-tolerant GPU virtualization mechanism that hooks CUDA APIs and uses Prometheus-based health monitoring to detect GPU and node failures and automatically migrate workloads to healthy GPUs or nodes, minimizing service disruption.

Quantitative Trading System Developer Outstanding Intern **Apr. 2022 – Sep. 2022**
ZhuoShi Private Equity Fund *Beijing, China*
• Develop an exchange's seat selection module based on the LambdaMart algorithm to select the seat with the lowest NDCG latency
• Develop a millisecond-level Kafka-based index calculation service

AWARDS & HONORS

ACM SIGHPC Student Travel Grant for SC 2026 Conference (6 spots, $1,900 USD)

RPg Research Performance Award of Hong Kong Baptist University ($20,000 HKD)

Merit Student of Dalian University of Technology

National Scholarship (Top 1%) of Dalian University of Technology

First Class Scholarship (Top 5%) of Dalian University of Technology

SKILLS

Techniques: C++, Cuda, Python

Languages: Mandarin(native), English(IELTS-7 CET 6-597)